EXP.NO:22

Print Intersection of 2 sorted arrays-(m*n) Time Complexity, O(1) Space Complexity

PROBLEM STATEMENT:

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

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CODE

```c
#include <stdio.h>
int main(){
    int a;
    scanf("%d",&a);
    for(int i=0;i<a;i++){
        int n;
        scanf("%d",&n);
        int arr[n];
        for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int m;
    scanf("%d",&m);
    int brr[m];
    for(int i=0;i<m;i++){
        scanf("%d",&brr[i]);
    }
    int c[m];
    int co=0;
    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            if(arr[i]==brr[j]){
                c[co++]=arr[i];
                break;
            }
        }
    }
    for(int i=0;i<co;i++){
        printf("%d ",c[i]);
    }
    }

}
```

OUTPUT:

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.